

Internet of Things

By

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Outline

- ▶ ML and Analytics in IoT
- ▶ The Strategic Value of IoT Data
- ▶ Anomaly detection
- ▶ Isolation Forests
- ▶ Properties of Isolation Forests
- ▶ Algorithm
- ▶ Advantage
- ▶ Practice code Isolation Forests

ML and Analytics in IoT

- ▶ IoT in Data Value Chain, Anomaly Detection, Predictive Maintenance, and Pattern Recognition.

ML and Analytics in IoT

- ▶ The true value of IoT is not in the "Thing," but in the **insights** derived from the data those things produce.
- ▶ **The Data Value Hierarchy:** We move from Descriptive (**What happened?**) to Diagnostic (**Why did it happen?**) to Predictive (**What will happen?**) and finally Prescriptive (**How can we make it happen?**).

The Strategic Value of IoT Data

- ▶ Unique Characteristics of IoT Data:
- ▶ **Time-Series Nature**: Almost all IoT data is stamped with a temporal coordinate.
- ▶ **High Velocity and Volume**: Sensors can fire every millisecond, creating massive datasets that require efficient preprocessing.
- ▶ **Noise and Sparsity**: IoT data is often "dirty" due to sensor degradation or network jitter.

The "Edge vs. Cloud" Trade-off

- ▶ Edge: Low latency, privacy-preserving, immediate action.
- ▶ Cloud: Long-term storage, heavy-duty model training, global pattern recognition.

Anomaly Detection in IoT

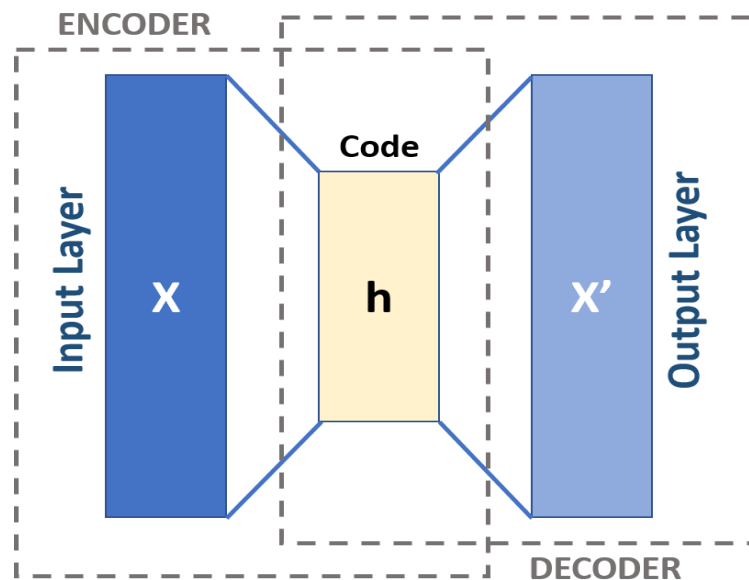
- ▶ Anomaly detection is the "first line of defense" in IoT, identifying data points that do not conform to expected behavior.
- ▶ Identifying cybersecurity breaches, sensor malfunctions, or sudden industrial equipment failures

ML Approaches:

- ▶ Statistical Methods: Using Z-scores or moving averages for simple thresholding.
- ▶ Unsupervised Learning: Since "normal" data is plentiful but "anomalous" data is rare.

Unsupervised learning

- ▶ **Isolation Forests**: Specifically designed to isolate anomalies rather than profiling normal points.
- ▶ **Autoencoders** : Training a neural network to reconstruct input data. High reconstruction error indicates an anomaly.

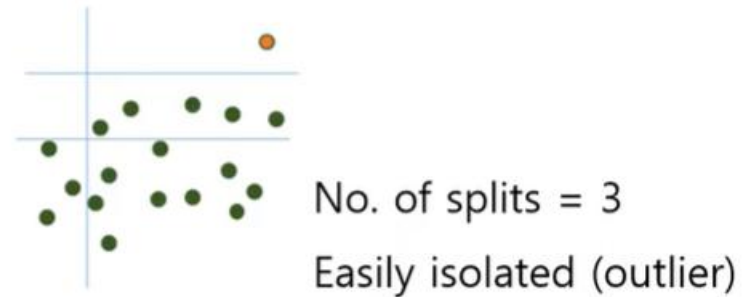
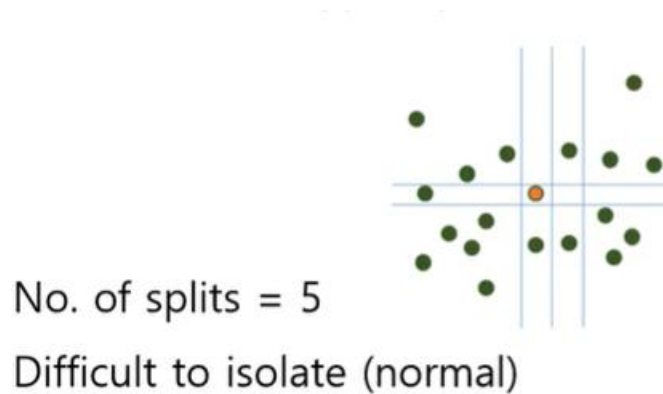


Predictive Maintenance

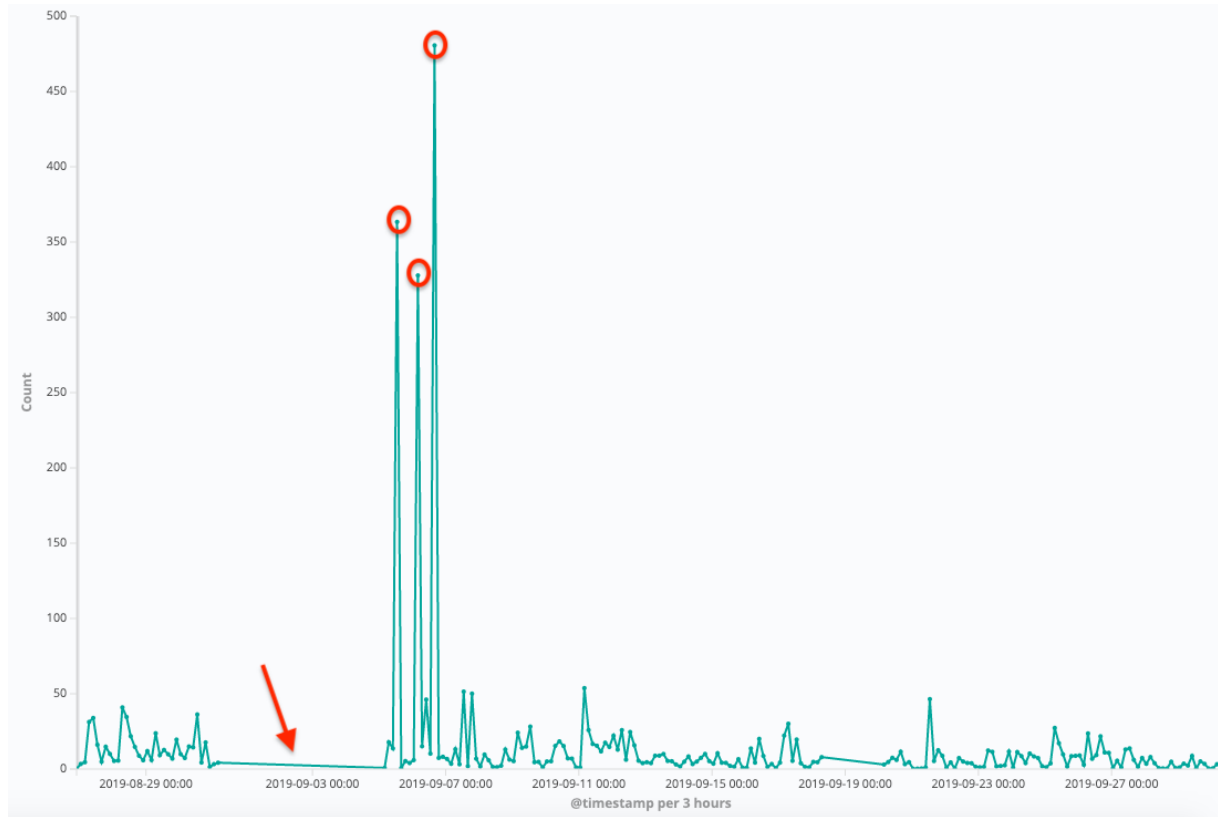
- ▶ PdM moves away from "Run-to-Failure" or "Fixed-Interval" maintenance toward a data-driven approach.
- ▶ The Objective: Predicting the Remaining Useful Life (RUL) of an asset.
- ▶ Modeling Techniques:
 - ▶ Regression Models: Predicting a continuous value (e.g., "This engine has 40 flight cycles left").
 - ▶ Classification: Predicting if a failure will occur within a specific window (e.g., "Will this pump fail in the next 7 days?").

Isolation Forests

An outlier detection algorithm. like sensor malfunctions, network intrusions, or unexpected environmental changes.



Isolation Forests



example web traffic with potentially anomalous points

Isolation Forests

- ▶ The graph in Fig.1 represents ingress traffic to a web server, expressed as the number of requests in 3-hours intervals, for a period of one month. It is quite evident by simply looking at the picture that **some points (marked with a red circle) are unusually high**, to the point of inducing suspect that the web server might have been under attack at that time. On the other hand, the flat segment indicated by the red arrow also seems unusual and might possibly be a sign that the server was down during that time period.

Isolation Forests

- ▶ Anomalies in a big dataset may follow very complicated patterns, which are difficult to detect "by eye" in the great majority of cases. This is the reason why the field of anomaly detection is well suited for the application of machine learning techniques.

Isolation Forests

- ▶ The most common techniques employed for anomaly detection are based on the construction of a profile of what is "normal": anomalies are reported as those instances in the dataset that do not conform to the normal profile .

Isolation Forests

- ▶ Isolation Forest uses a different approach: instead of trying to build a model of normal instances, it explicitly isolates anomalous points in the dataset. The main advantage of this approach is the possibility of exploiting sampling techniques to an extent that is not allowed to the profile-based methods, creating a very fast algorithm with a low memory demand.

Isolation Forests

- ▶ The Isolation Forest (iForest) algorithm was initially proposed by Fei Tony Liu, Kai Ming Ting and Zhi-Hua Zhou in 2008.
- ▶ The authors took advantage of two quantitative properties of anomalous data points in a sample, that is:
 - ▶ 1) they are the minority consisting of fewer instances and
 - ▶ 2) they have attribute-values that are very different from those of normal instances

Isolation Forests

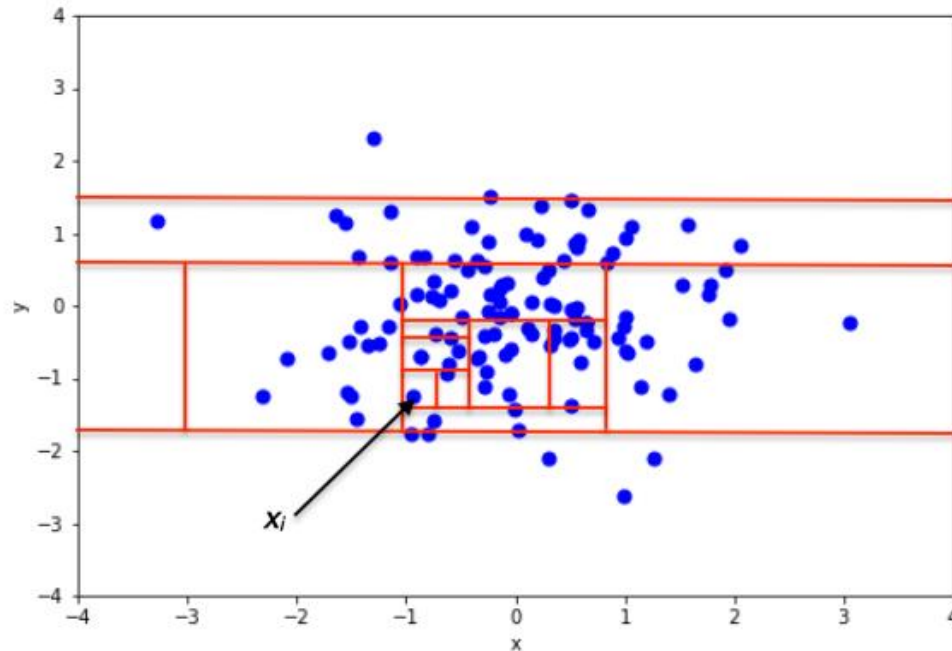
- ▶ Since anomalies are typically few and very different from the other points in the sample, they must be easier to "isolate" compared to normal points. On the basis of this principle, Isolation Forest builds an ensemble of "Isolation Trees" (iTrees) for the data set and marks as anomalies the points that have short average path lengths on the iTrees.

Advantage

- ▶ has a low linear time complexity and a small memory requirement
- ▶ is able to deal with high dimensional data with irrelevant attributes
- ▶ can be trained with or without anomalies in the training set
- ▶ can provide detection results with different levels of granularity without re-training

Isolation Forests

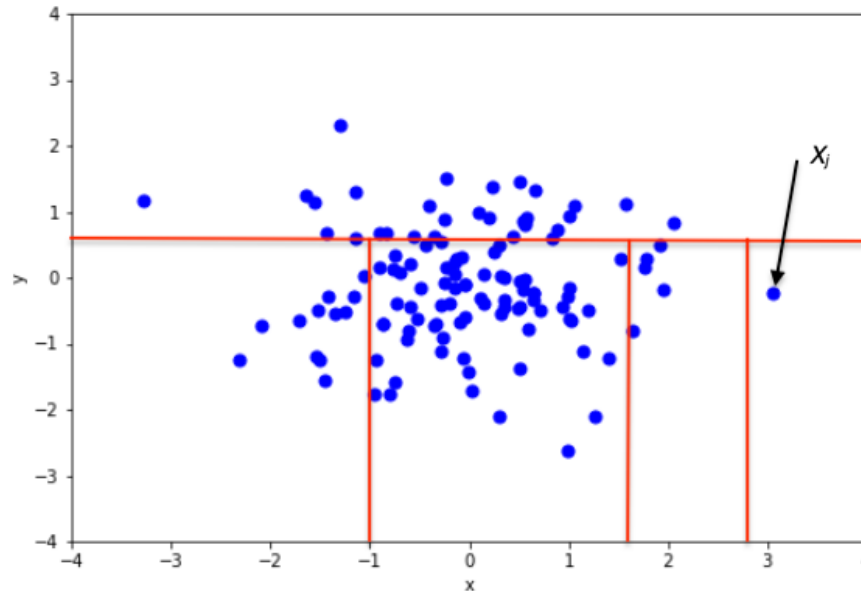
In order to isolate a data point the algorithm recursively generates partitions on the sample by randomly selecting an attribute and then randomly selecting a split value for the attribute, between the minimum and maximum values allowed for that attribute.



an example of isolating a non-anomalous point in a 2D Gaussian distribution

Isolation Forests

- ▶ An example of random partitioning in a 2D dataset of normally distributed points is given in Fig. 2 for a non-anomalous point and Fig. 3 for a point that's more likely to be an anomaly. It is apparent from the pictures how anomalies require fewer random partitions to be isolated, compared to normal points.



Isolation Forests

- ▶ From a mathematical point of view, recursive partitioning can be represented by a tree structure named *Isolation Tree*, while the number of partitions required to isolate a point can be interpreted as the length of the path, within the tree, to reach a terminating node starting from the root. For example, the path length of point x_i in Fig. 2 is greater than the path length of x_j in Fig. 3.

Isolation Forests

- ▶ let $X = \{x_1, \dots, x_n\}$ be a set of d -dimensional points and $X' \subset X$ a subset of X . An Isolation Tree (iTree) is defined as a data structure with the following properties:
- ▶ for each node T in the Tree, T is either an external-node with no child, or an internal-node with one "test" and exactly two daughter nodes (T_l, T_r)
- ▶ a test at node T consists of an attribute q and a split value p such that the test $q < p$ determines the traversal of a data point to either T_l or T_r .

Properties of Isolation Forest

- ▶ **Sub-sampling:** since iForest does not need to isolate all of normal instances, it can frequently ignore the big majority of the training sample. As a consequence, **iForest works very well when the sampling size is kept small**, a property that is in contrast with the great majority of existing methods, where large sampling size is usually desirable.
- ▶ **Swamping:** **when normal instances are too close to anomalies, the number of partitions required to separate anomalies increases, a phenomena known as *swamping***, which makes it more difficult for iForest to discriminate between anomalies and normal points. One of the main reasons for swamping is the presence of too many data for the purpose of anomaly detection, which implies one possible solution to the problem is sub-sampling. Since iForest respond very well to sub-sampling in terms of performance, the reduction of the number of points in the sample is also a good way to reduce the effect of swamping.

Properties of Isolation Forest

- ▶ **Masking:** when the number of anomalies is high it is possible that some of those aggregate in a dense and large cluster, making it more difficult to separate the single anomalies and, in turn, to detect such points as anomalous. Similarly to swamping, this phenomena (know as "*masking*") is also more likely when the number of points in the sample is big, and can be alleviated through sub-sampling

Properties of Isolation Forest

- ▶ **High Dimensional Data:** one of the main **limitation** to standard, **distance-based methods** is **their inefficiency in dealing with high dimensional datasets**: The main reason for that is, in a high dimensional space every point is equally sparse, so using a **distance-based measure of separation is pretty ineffective**. Unfortunately, high-dimensional data also affects the detection performance of **iForest**, but the performance can be vastly improved by adding a features selection test like Kurtosis to reduce the dimensionality of the sample space.
- ▶ **Normal Instances Only:** iForest performs well even if the training set does not contain any anomalous point, the reason being that iForest describes data distributions in such a way that **high values of the path length $h(x_i)$ correspond to the presence of data points**. As a consequence, the presence of anomalies is pretty irrelevant to iForest's detection performance.

Anomaly Detection with Isolation Forest

- ▶ Two main stages:
- ▶ in the first stage, a training dataset is used to build iTrees
- ▶ in the second stage, each instance in test set is passed through the iTrees build in the previous stage, and a proper "anomaly score" is assigned to the instance using the algorithm (BST).

Anomaly score

- ▶ a termination to an external node of the iTree corresponds to an unsuccessful search in the BST. As a consequence, the estimation of average $h(x)$ for external node terminations is the same as that of the unsuccessful searches in BST, that is

$$c(m) = \begin{cases} 2H(m-1) - \frac{2(m-1)}{n} & \text{for } m > 2 \\ 1 & \text{for } m = 2 \\ 0 & \text{otherwise} \end{cases}$$

where n is the testing data size, m is the size of the sample set and H is the harmonic number, which can be estimated by

$$H(i) = \ln(i) + \gamma, \text{ where } \gamma = 0.5772156649 \text{ is the Euler-Mascheroni constant.}$$

Anomaly score

- ▶ The value of $c(m)$ above represents the average of $h(x)$ given m , so we can use it to normalise $h(x)$ and get an estimation of the anomaly score for a given instance x :

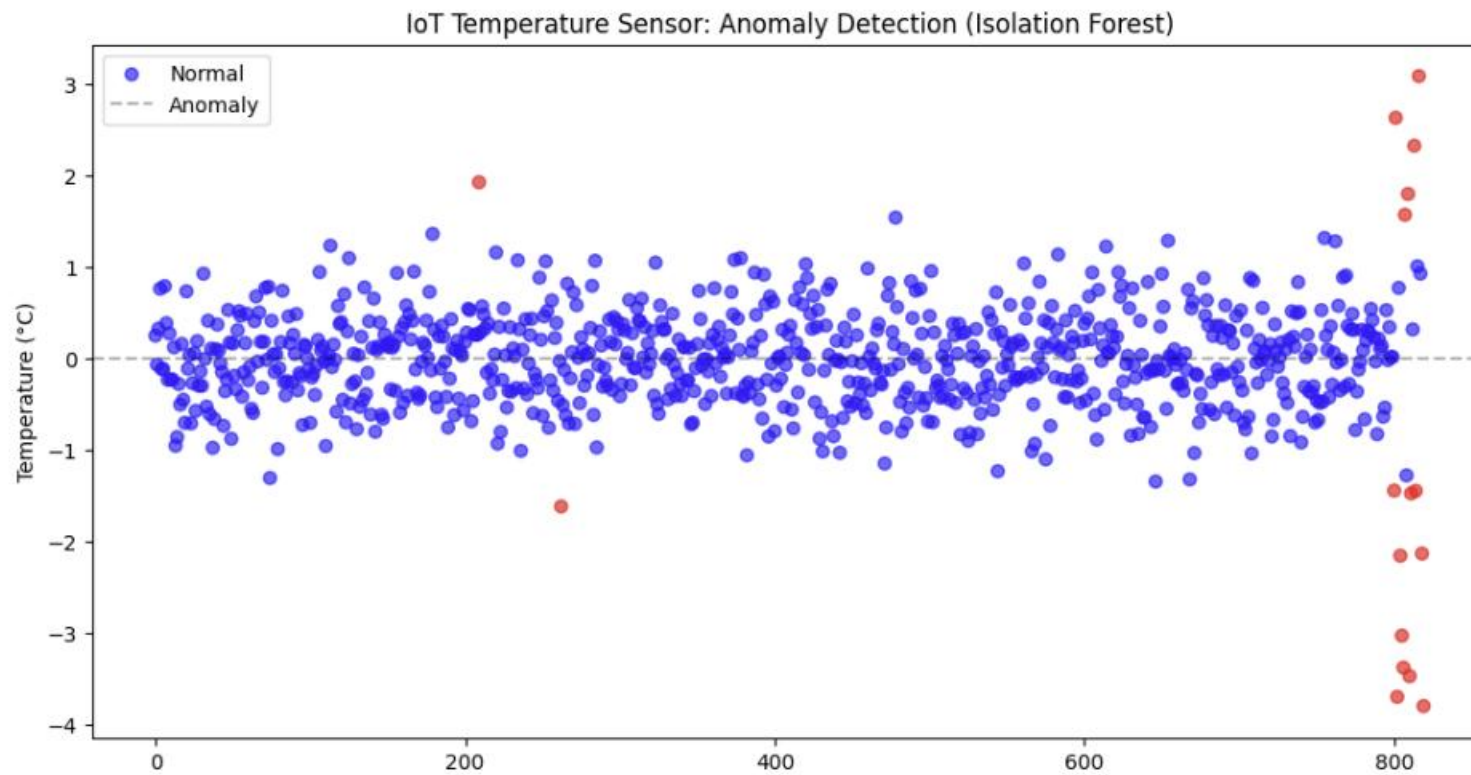
$$s(x, m) = 2 \frac{-E(h(x))}{c(m)}$$

- ▶ where $E(h(x))$ is the average value of $h(x)$ from a collection of iTrees. It is interesting to note that for any given instance x :
- ▶ if s is close to 1 then x is very likely to be an anomaly
- ▶ if s is smaller than 0.5 then x is likely to be a normal value
- ▶ if for a given sample all instances are assigned an anomaly score of around 0.5, then it is safe to assume that the sample doesn't have any anomaly

Practice code

- ▶ We are monitoring a sensor that typically reports temperatures between -2°C and 2°C . We will simulate:
- ▶ **Normal Data:** Steady fluctuations within the expected range.
- ▶ **Anomalies:** A sudden spike (door left open) and a sensor "drift" (hardware failure).

Practice code



Practice code

- ▶ include a second sensor : Humidity
- ▶ Generate 800 normal readings for Humidity between 40% and 60%.
- ▶ Add 20 anomalies where Humidity drops suddenly to 10%.
- ▶ Update the IsolationForest to train on both Temperature and Humidity

Reference

- ▶ https://en.wikipedia.org/wiki/Isolation_forest
- ▶ <https://cs.stanford.edu/~quocle/tutorial2.pdf>

Question



Thank you

